

Lecture 06

High Level Hardware Synthesis I

Descriptive Question

1. Explain High Level Synthesis design steps with the help of a block diagram.
2. Define Data flow graph with suitable example.
3. Describe **ASAP scheduling** in detail using a suitable example and diagram.
4. Describe **ALAP scheduling** in detail and explain how it is computed.
5. Describe the goal and approach of Resource constraints scheduling.
6. Describe the goal and approach of Time constraints scheduling.
7. Explain **resource-constrained scheduling** and list scheduling with a detailed example.
8. Discuss **time-constrained scheduling** and explain why force-directed scheduling is used.
9. Explain the construction of a **Distribution Graph (DG)** step by step.
10. Describe the **force calculation formula** used in force-directed scheduling and explain each term.
11. Explain the **TC Scheduling Algorithm** step by step.
12. Compare **ASAP, ALAP, resource-constrained, and time-constrained scheduling** in tabular form.
13. Explain how force-directed scheduling minimizes hardware resources under fixed time constraints.
14. Distinguish between FPGA and ASIC.
15. Define a critical path in scheduling.
16. Define mobility (movability) of an operation.

Mathematical Question

1. Consider the following algorithm:

(i) Algorithm: $x1 := x + dx;$ $y1 := y + (u * dx);$ $u := u - (5 * x * u * dx) - (3 * y * dx);$	(ii) Algorithm: $1) t1 = a + b$ $2) t2 = c + d$ $3) t3 = t1 * t2$ $4) y = t3 - e$
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- (a) Draw the Data Flow Graph (DFG) for the given algorithms.
 (b) Perform **ASAP scheduling** for the above algorithm and determine the total latency.
 (c) Perform **ALAP scheduling** assuming a latency of 4 cycles.
 (d) Calculate the **mobility** of each operation:

$$Mobility = ALAP - ASAP$$

2. Consider the following algorithm and resource constraints:

(i) Algorithm: $1) p = a * b$ $2) q = c * d$ $3) r = p + q$ $4) s = r - e$ $5) y = s + f$ Resource Constraints: • 1 multiplier • 1 ALU (+/-)	
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- (a) Draw the **DFG**.
 (b) Perform **resource-constrained list scheduling** (show cycle-by-cycle table).
 (c) Find:
 (i) total latency (number of cycles)
 (ii) which operations execute in each cycle
 (d) Compare with **ASAP scheduling** (no resource limit). Determine which is faster and why.

3. Consider the following algorithm and time constraint:

(i) Algorithm: 1) $t1 = a + b$ 2) $t2 = t1 + c$ 3) $t3 = d * e$ 4) $t4 = t3 * f$ 5) $y = t2 + t4$ Time Constraints: • Must finish in 4 cycles	
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(a) Draw the **DFG**.

(b) Compute **ASAP** and **ALAP** schedules for latency = 4.

(c) For each operation, compute its **possible cycles range** (mobility window).

(d) Build **DG for ADD** and **DG for MULT** using rule:

If an operation can be scheduled in **k cycles**, each cycle gets weight: $\frac{1}{k}$.

(e) Plot or tabulate DG values per cycle.

4. Consider the following algorithm and time constraint:

(i) Algorithm: 1) $t1 = a * b$ 2) $t2 = c * d$ 3) $t3 = t1 + t2$ 4) $t4 = t3 * e$ 5) $y = t4 + f$ Time Constraints: • 4 cycles	
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Given DGmult values (after initial DG):

cycle	DGmult
1	0.8
2	1.2
3	1.4
4	1.0

Assume operation $t4 = t3 * e$ (a multiplication) has mobility window which can be scheduled in cycle 3 **or** 4. So, $\Delta T = 2$.

(a) Compute avg DGmult over its possible cycles.

(b) Compute

(i) $Force(3)$

(ii) $Force(4)$

(c) Determine the best cycle to schedule it and explain using **minimum force rule**.

5. Consider the following algorithm and time constraint:

(i) Algorithm: 1) $t1 = a + b$ 2) $t2 = c + d$ 3) $t3 = t1 * t2$ 4) $t4 = t3 + e$ 5) $y = t4 * f$ Time Constraints: • 5 cycles	
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- (a) Perform **ASAP** and **ALAP** scheduling (latency = 5).
- (b) Build DG for:
 - (i) ADD operations (1,2,4)
 - (ii) MULT operations (3,5)
- (c) Choose **one unscheduled operation** that has mobility > 0 (not fixed).
- (d) Compute force for that operation for each possible cycle.
- (e) Select the cycle with minimum force (first decision of TC scheduling).
- (f) Update DG after that assignment (show updated DG values).

(Practice the questions discussed in the class Lecture)